

JOSHUA COWELL

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EDUCATION

Bachelor of Science in Mechanical Engineering

Expected May 2027

Liberty University, Lynchburg, VA

GPA: 3.82

Relevant Coursework: Fluid Dynamics, Heat Transfer, Thermodynamics I & II, Differential Equations, Dynamics, Materials Engineering, Statics, Calculus I–III

PROFESSIONAL EXPERIENCE

Thermal Hydraulics Engineering Intern

May 2025 – Dec 2025

Framatome Nuclear

- Developed a predictive wear model in Python for safety-critical reactor components, achieving 75% improvement in accuracy over legacy methods
- Engineered a Python algorithm processing 6-dimensional datasets to calculate safe operating limits, reducing analysis time from 800+ hours to minutes
- Performed system-level thermal-hydraulic and flow-induced vibration (FIV) analysis to support component qualification
- Authored technical documentation and presented findings to engineering leadership; awarded \$2,500 for presentation excellence

Chief Systems Engineer

June 2025 – Present

Liberty Rocketry (9th of 156 international teams, 2024)

- Lead a cross-functional team of 60+ members in the design, integration, and verification of high-powered rocket systems
- Direct systems engineering decisions across propulsion, avionics, recovery, and aerodynamics subsystems
- Manage requirements definition, interface control, and configuration management throughout the development lifecycle
- Oversee a \$16,000+ project budget, vendor coordination, and schedule milestones for competition campaigns
- Perform verification and validation (V&V) using SolidWorks, ANSYS, and MATLAB for subsystem models

Lead Undergraduate Research Assistant

Sep 2024 – Present

Liberty University

- Lead a team of 5 researchers conducting split-Hopkinson pressure bar (SHPB) testing for dynamic material characterization
- Design test procedures and fixtures for high-strain-rate experiments; analyze results using electron microscopy and high-speed imaging

Senior Structural Engineer

Sep 2023 – June 2025

Liberty Rocketry

- Conducted finite element analysis (FEA) and computational fluid dynamics (CFD) simulations to verify structural integrity and aerodynamic performance
- Authored technical reports documenting analysis methodology, results, and compliance with competition requirements

Camp Counselor

May 2024 – Aug 2024

Woodlands Camp, GA

- Provided 24-hour supervision during weeklong outdoor expeditions with 12 youth (ages 8–18), instructing survival skills, rock climbing, whitewater rafting, and cave camping

TECHNICAL SKILLS

Analysis & CAE: ANSYS Fluent, ANSYS Mechanical, AFT Fathom, OpenRocket, RocketPy, FEA, CFD

CAD & Design: SolidWorks, Onshape, Technical Drawing, 3D Printing, CNC Machining

Programming: Python (NumPy, Xarray, Pandas), MATLAB, Data Analysis, Automation

Laboratory: Scanning Electron Microscope (SEM), High-Speed Imaging, Material Testing

PROJECTS

Predictive Wear Model – Nuclear Component Lifecycle Analysis

- Developed a Python-based predictive model for safety-critical components, improving failure prediction accuracy by 75%

Automated Operating Limit Analysis Tool

- Engineered a Python automation framework reducing 800+ hours of manual engineering calculations to minutes

Heat Exchanger Thermal-Hydraulic Optimization

- Performed CFD analysis using ANSYS Fluent to optimize heat exchanger thermal performance and flow characteristics

Unmanned Aerial Vehicle (UAV) Design & Analysis

- Designed a modular UAV airframe with rapid-prototyping capability; validated aerodynamics through CFD simulation

3D Printed High-Power Rocket

- Engineered a fully 3D printed rocket achieving Mach 0.5+ and 3,000 ft altitude; performed ANSYS CFD analysis for aerodynamic optimization

High Strain Rate Bone Research

- Investigating the dynamic loading behavior of human bone using Split Hopkinson Pressure Bar for conference presentation